

Scientific Computation

Prof. Matthew Giamou
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Lecture 5 - The Taylor Series

Summary

Three ingredients for cancellation to occur:

Ariane 5

Recall the disaster video from the first lecture - the June 4th, 1996 launch of the Ariane 5 rocket.

A 64-bit floating point number was converted to a 12-bit integer!
Overflow checking was limited for performance reasons, and assumptions from the Ariane 4's flight were incorrectly made



Vancouver Stock Exchange

- In 1982, the Vancouver Stock Exchange started an electronic stock index set initially to 1000 points
- The index was updated after each transaction
- In 22 months the index **fell to 520**
- It was not supposed to fall in a bull market!
- Investigation showed each intermediate result was rounded to 2 **decimals** by chopping, e.g., 568.958 rounds to 568.95
- Once corrected, the index was at **1098.892**

Lecture Outline

- **Taylor series refresher**
- Truncation vs. roundoff error
- Computational error

The Mean Value Theorem

If a function f has n continuous derivatives in $[a, b]$, we say $f \in C^n[a, b]$.

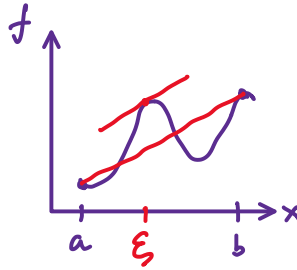
continuous function



Mean Value Theorem: If $f \in C^0[a, b]$, $a < b$, then

$$f(b) = f(a) + (b - a)f'(\xi) \text{ for some } \xi \in (a, b).$$

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}$$



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Taylor Series

↗ $f \in C^\infty[a, b]$

Taylor series of an infinitely differentiable (real or complex) function f at c :

$$f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \dots$$

$$= \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!} (x-c)^k$$

Expand at $x = c$

Maclaurin series : Taylor series $c = 0$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!} x^2 + \dots$$
$$= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$$

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Taylor's Theorem

Assume f has $n + 1$ continuous derivatives in $[a, b]$, which we denote

$f \in C^{n+1}[a, b]$. Then for any c and x in $[a, b]$:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(c)}{k!} (x-c)^k + E_{n+1}$$

approx Error

$$\text{where } E_{n+1} = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-c)^{n+1}$$

where $\xi = \xi(x, c)$, $\xi \in [c, x]$ where $x > c$
 $\xi \in [x, c]$ where $c > x$

ξ is a function of x and c .

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Error Term

So at x to expand, and h is step size to control the error

Replacing x by $x + h$ and c by x , we obtain

...

$$f(x+h) = \sum_{k=0}^n \frac{f^{(k)}(x)}{k!} h^k + E_{n+1}$$

$$\text{where } E_{n+1} = \frac{f^{(n+1)}(\xi)}{(n+1)!} h^{n+1}, \quad \begin{array}{l} \xi \in [x, x+h] \\ \text{if } h > 0, \xi \in [x+h, x] \text{ otherwise} \end{array}$$

We say that the error term E_{n+1} is of order $n+1$, and

we write $E_{n+1} = O(h^{n+1})$, that is $|E_{n+1}| \leq \alpha h^{n+1}$ for some $\alpha > 0$

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Approximating e^x

Example 1: How can we approximate e^x for a given x ?

$$f^{(k)}(x) = f(x) \text{ for } e^x$$

$$e^x = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = 1 + x + \frac{x^2}{2!} + \dots$$

$$\text{suppose we use } e^x \approx 1 + x + \frac{x^2}{2} \quad (k=2)$$

$$e^x = 1 + x + \frac{x^2}{2} + E_3, \text{ where } E_3 = \frac{e^{\xi}}{3!} x^3, \quad \xi \in [0, x]$$

$$\text{Let } x = 0.1, \text{ then } e^{0.1} \approx 1.1052, \text{ The error is } E_3 = \frac{e^{\xi}}{3!} 0.1^3 \leq \frac{e^{x=0.1} \cdot 0.1^3}{3!} \\ = 1.842 \times 10^{-4}$$

Checking our Approximation of e^x

Example 2: How can we check our calculation? Let's compute a more accurate value using Julia's BigFloat function:

```
x = 0.1
approx_error = exp(BigFloat(x)) - (1 + x + x^2/2)
✓ 1.2s
0.00017091807564764871020721746128430451977010582734462084148346131070925705336

error_bound = exp(x)*x^3/(factorial(3))
✓ 0.1s
0.000184195153012608

approx_error <= error_bound
✓ 0.1s
true
```

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Adding Another Term for e^x

Example 3: Let's approximate e^x using a third term: $x = 0.1$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}, \text{ the error is } E_4 = \frac{e^3}{4!} x^4 \leq \frac{e^{0.1}}{4!} (0.1)^4 \approx 4.605 \times 10^{-6}$$

↳ Repeat the coding exercise from the previous slide:

$$\exp(x) - \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}\right)$$

↓ $\approx 4.25 \times 10^{-6} < \epsilon_4$
computed error of our approximation

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Lecture Outline

- Taylor series refresher
- **Truncation vs. roundoff error**
- Computational error

Roundoff Errors

Example 4: Consider computing $\exp(0.1)$. The binary representation of 0.1 is infinite:

binary
↗

$$(0.1)_{\beta=10} = (0.0\ 0011\ 0011\ldots)_{\beta=2}$$

- In FP, some common decimal number that don't repeat have an infinite repeating representation that gets rounded
- The input to $\exp(0.1)$ already has round off error
- The $\exp()$ function has its own error
- The output of $\exp(0.1)$ is also rounded
(round to decimal again)

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Truncation Errors

$$\text{Consider } e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \boxed{\sum_{k=4}^{\infty} \frac{x^k}{k!}}.$$

Suppose we approximate:

$$e^x \approx 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}, \text{ that is we truncate the series}$$

~~X~~ The resulting error (E_4) is a **truncation error**

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Approximating the First Derivative

$$f \in C^2[a, b]$$

Let $f(x)$ be a function with continuous second derivative.

$$f(x+h) = f(x) + f'(x)h + \frac{f''(\xi)h^2}{2!}, \quad \xi \in [x, x+h]$$

$$f'(x) = \frac{f(x+h) - f(x)}{h} - \frac{f''(\xi)h}{2!}$$

Therefore, if we use $f'(x) \approx \frac{f(x+h) - f(x)}{h}$
then the truncation error is $-\frac{f''(\xi)h}{2!}$

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Computational Error

$$\text{Computational error} = (\text{rounding error}) + (\text{truncation error})$$

Round off

Rounding error: difference between the result produced using finite-precision arithmetic (e.g., IEEE Float32) and exact arithmetic

Truncation error: difference between the true result and the result that would be produced by an (inexact) *algorithm* using exact arithmetic

↳ e.g. Taylor series cut off

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2nd Order Truncation Error

Example 5: Approximate $f'(x)$ with $\frac{1}{h}(f(x+h) - f(x))$.

$\tau = -\frac{1}{2}f''(\xi)h$ for $\xi \in [x, x+h]$ last ex.

→ Notice no mention of FP or rounding error

Computational Error Example

Example 6 cont.: The we approximate $f'(x)$ by:

$$\frac{f_1 - f_2}{h} = \frac{f(x+h) - f(x)}{h} + \frac{\delta_1 - \delta_2}{h}$$

Ignoring the error in the subtraction and division in $(f_1 - f_2)/h$, the total computational error is:

$$\begin{aligned} \underbrace{f'(x)}_{\text{true value}} - \frac{f_1 - f_2}{h} &= \frac{f(x+h) - f(x)}{h} - \frac{1}{2} f''(\xi) h - \frac{f(x+h) - f(x)}{h} - \frac{\delta_1 - \delta_2}{h} \\ &= -\frac{1}{2} f''(\xi) h - \frac{\delta_1 - \delta_2}{h} \end{aligned}$$

Denote by M the max of $|f''(x)|$ for $x \in [x, x+h]$
we assume $|\delta_1|, |\delta_2| \leq \varepsilon_{mach}$

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Computational Error Example

Example 6 cont.: Then

$$\left| f'(x) - \frac{f_1 - f_2}{h} \right| = \left| -\frac{1}{2} f''(\xi) h - \frac{\delta_1 - \delta_2}{h} \right|$$

$$\leq \left| \frac{1}{2} f''(\xi) h \right| + \left| \frac{\delta_1 - \delta_2}{h} \right|$$

$$\leq \left| \frac{1}{2} M h \right| + \frac{|\delta_1| + |\delta_2|}{h}$$

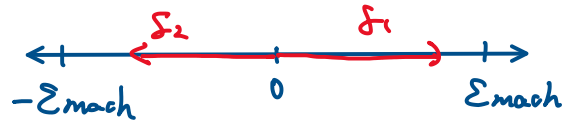
triangle
inequalities

M is max of $|f''(x)|$ for
 $x \in [x, x+h]$, and $|\delta_1|,$

$$\leq \frac{1}{2} M h + \frac{2\varepsilon_{mach}}{h}$$

$$|\delta_2| \leq \epsilon_{mach}$$

$$|\delta_1| \leq \epsilon_{mach}, \quad |\delta_2| \leq \epsilon_{mach}$$



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Note: 这里为了推导方便，把 $f(x+h)$ 和 $f(x)$ 到FP形式的误差表示为了加法形式，即绝对误差，其实也可以写成之前那种相对误差但是推导麻烦。

注意，不管是泰勒余项还是roundoff error都是absolute error。